Faculty of Computing & Information Technology

Course: Database Management Systems (DBMS)

Laboratory Manual - Experiment 08

Title: Relational Library Schema & Circulation Tracking

1. Relational Schema Specification

Books (BookID INT PRIMARY KEY, Title VARCHAR(100), Author VARCHAR(100), Category VARCHAR(50), Price DECIMAL(10,2))

Members (MemberID INT PRIMARY KEY, FullName VARCHAR(100), Email VARCHAR(100), JoinDate DATE)

Loans (LoanID INT PRIMARY KEY, BookID INT REFERENCES Books(BookID), MemberID INT REFERENCES Members(MemberID), LoanDate DATE, ReturnDate DATE)

2. LAB TASKS:

Query 1: Relational Table Creation with Constraints: Write SQL statements to create Books, Members, and Loans tables with PRIMARY KEY and FOREIGN KEY constraints.

Query 2: Book Circulation Category Aggregation: Write a query grouping books by Category, counting total books and computing average price, displaying only categories with at least 1 book.

Query 3: Member Active Borrowing Join Report: Write an INNER JOIN query between Members and Loans to retrieve member names, book IDs, and loan dates for active borrowings.

3. Analysis Questions:

1. How do foreign key constraints enforce referential integrity across the Books and Loans tables?

2. Compare the query execution plan of an INNER JOIN versus subquery with IN clause.

4. Viva Voce Questions:

1. What is the fundamental difference between WHERE and HAVING clauses in SQL?

2. Can a table have multiple foreign keys referencing different tables?